



ELECT MAINT- MAIN PLANT Department
SIMHADRI

Note No :1



Ref:SMTTP/ELECT MAINT-/2023-24/EMD CHP/373099

Date:07/08/2023(16:17:09)

Subject:New LMI on Guidelines for Safety in Electrical systems

New OGN No. COS-OGN/OPS/ELEC/043, Rev. No: 0 June 2023 has been issued on " Guidelines for Safety in Electrical systems", copy is attached herewith.

Based on this OGN, an LMI no. LMI/OGN/OPS/ELEC/043 is prepared for electrical safety pertaining to Simhadri. LMI copy is attached herewith.

Put up for approval please.

Submitted by:

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Note No:2

Please route through all sectional incharges of EMD for comprehensive Review.

Submitted by:

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Note No:3

Plz review

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Note No:9

Reviewed and Read the subject LMI

Submitted by:

सुर्या प्रकाश राव पसुपुरेड्डी - वरिष्ठ प्रबंधक

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Note No:10

Reviewed the LMI as per OGN.

May process further and approve

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Note No:11

New LMI on guidelines for electrical safety system reviewed

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Note No:12

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Note No:13

please add 2 years review clause in LMI

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Note No:14

Review cluase included at sl.no.6, and the document is updated.

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Note No:15

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Note No:16

Pl examine

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Note No:17

Reviewed and Read the subject LMI

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Note No:18

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Note No:20

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Note No:21

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Note No:24

Please review

Submitted by:

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Note No:25

LMI point 9.19, 9.32, 9.33 to be reviewed as per OGN.

LMI point 10.1 Table 2, As per OGN value for - Height of Insulator bottom from Ground (mm) is 2440 mm.
As per LMI value is 2400 mm. May be reviewed.

LMI point 10.4.3 , point 4 and 5 may be reviewed.

Submitted by:

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Note No:26

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Note No:27

Points at Note no. 25 are modified in the LMI.

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Note No:28

LMI reviewed. Reviews suggested in Note No-25 are in order.

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Note No:29

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Note No:30

Reviewed and submitted for approval

Submitted by:

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Note No:31

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Note No:32

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Note No:33

Reviewed and read the subject LMI

Submitted by:

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Note No:34

Please review.

Submitted by:

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Note No:35

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Note No:36

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Note No:37

LMI is reviewed and found in line with OGN.

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Note No:38

LMI forwarded for further approval (All Sections have reviewed the LMI and corrections suggested by Sr Manager(MM-AHP) incorporated.

Submitted by:

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Note No:39

May please be accorded approval.

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Note No:40

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Note No:41

Approved & Submitted by:

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NTPC LIMITED

SIMHADRI THERMAL POWER PLANT

ELECTRICAL MAINTENANCE DEPARTMENT

LOCATION MANAGEMENT INSTRUCTION

LMI Name:	Guidelines for Safety in Electrical systems	
LMI No.:	LMI/OGN/OPS/ELEC/043	
Issue No.: 1	Revision No.: 0	Date: Aug'2023

Online Approval for Implementation by:

Sanjay Kumar Sinha, HOP (SIMHADRI)

E Office Ref No.: SMTTPP/ELECT MAINT-/2023-24/EMD CHP/373099

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1.0 INTRODUCTION

Electricity is widely recognized as a serious workplace hazard, exposing employees to electric shock, burns, fires, and explosions. It is well known that the human body will conduct electricity. If direct body contact is made with an electrically energized part while a similar contact is made simultaneously with another conductive surface that is maintained at a different electrical potential, a current will flow, entering the body at one contact point, traversing the body and then exiting at the other contact point, usually the ground. Each year many persons suffer pain, injuries and death from such electric shocks.

Associated Electrical hazards are Electrical Shock, Electrical Flash, Electrical Burn, Fire, Slip/Trip/ Fall, Hit/ Press/ /Cut hazard, Fall of Person or materials from height etc. The objective of this OGN is to provide a generic guideline for electrical safety practices to be observed in NTPC generation stations.

2.0 SUPERSEDED DOCUMENT

Nil.

3.0 RECORDS OF REVISION

Nil.

4.0 REFERENCE DOCUMENT

COS-OGN/OPS/ELEC/043 Rev. No. 0, July 2023

5.0 SCOPE

All electrical installations of Simhadri Power Plant including Floating Solar power installation.

6.0 REVIEW

6.1 Review Responsibility : AGM I/c- EMD, Simhadri.

6.2 Review Frequency : 2 yearly basis or as and when required, whichever is earlier

6.3 Review Approval : HoP (Simhadri)

7.0 ELECTRICAL SHOCK:

Burns suffered in electrical accidents can be very serious. These burns may be of three basic types: electrical burns, arc burns, and thermal contact burns. Electrical burns are the result of the electric current flowing in the tissues and may be either skin deep or may affect deeper layers (such as muscles and bones) or both. Tissue damage is caused by the heat generated

from the current flow; if the energy delivered by the electric shock is high, the body cannot dissipate the heat, and the tissue is burned. Typically, such electrical burns are slow to heal.

Arc burns are the result of high temperatures produced by electric arcs or by explosions close to the body. Finally, thermal contact burns are those normally experienced from the skin contacting hot surfaces of overheated electric conductors, conduits, or other energized equipment.

It is well known that the human body will conduct electricity. Current through the body, even at levels as low as 3 milliamperes, can also cause injuries of an indirect or secondary injuries in which involuntary muscular reaction from the electric shock can cause bruises, bone fractures and even death resulting from collisions or falls. Table for body current flow Vs affect on person is shown here:

Current flow (mA)	Affect on Person
0.5-3mA	Tingling sensations
3-10mA	Muscle contractions and pain
10-40mA	"Let-go" threshold
30-75mA	Respiratory paralysis
100-200mA	Ventricular fibrillation
200-500mA	Heart clamps tight
>1500mA	Tissue and Organs start to burn

8.0 THE HIERARCHY OF HEALTH & SAFETY CONTROLS:

MOST EFFECTIVE ↓ ↓ ↓ ↓ LEAST EFFECTIVE	Elimination or Substitution	• Substitute with less hazardous material • Change process to eliminate noise • Perform task at ground level • Automated material handling
	Engineering Controls	• Ventilation systems • Machine guarding • Electrical enclosures • Current limiting circuit breakers/fuses • Interlocks • Lift tables, conveyors, balancers
	Warnings	• Signs • Barricades • Horns • Labels
	Training & Procedures	• Safe job procedures • Safety equipment inspections • Hazard Communications Training • Lock-out • Confined Space Entry, etc...
	Personal Protective	• Safety glasses • Ear plugs
	Equipment	• Flame Retardant Clothing • Face shields • Safety harnesses and lanyards • Safety shoes & Gloves

Research indicates that fixing the workplace approach is actually more effective and less expensive in the long run. One reason is that human behavior can never be completely regulated and controlled, so solutions based on compliance with procedures will always lead to mishaps. Control of Hazardous Energy (LOTO) is much more capable of guaranteeing safety and health. The UAW's and OSHA's analysis of control effectiveness is captured in the graph below.

9.0 GENERAL SAFETY GUIDELINES IN ELECTRICAL SYSTEMS:

Electrical work can be performed safely if the rules are followed, right precautions are taken and one stay aware of the surroundings. All Electrical Installations shall be installed and maintained in compliance with the Central Electricity Authority (Measures relating to Safety and Electricity Supply) Regulations.

Following general guidelines to be followed by the electrical systems including all the involved manpower:

- 1) It must be ensured that all safety practices are to be adhered as per Safety policy & rules mentioned in NTPC safety manual (Yellow book) & NTPC E&M safety rules.
- 2) It must be ensured that all the activities pertaining to the Electrical system shall be carried out in supervision of concern area employee with proper PTW & PPEs. Also, SOP to be adhered during any switching operation in Switchyard & Switchgears.
- 3) The Electrical Installations for Medium, High and Extra high voltage shall have a danger notice both in English or Hindi and local language of the district with a sign of skulls and bones.
- 4) Minimum PPE requirement, Hazards & precautions to be taken, Unauthorized Access Prohibited Signage, Schematic layout of the Electrical Area with emergency Equipment location etc. are to be displayed at the entrance of Electrical Room.
- 5) Electrical Shock Chart with instructions in English or Hindi and the local language of the district and where Hindi is the local language, in English and Hindi for the restoration of persons suffering from electric shock have to be affixed in a conspicuous place.
- 6) Single Line Diagram to be displayed inside Electrical rooms.
- 7) The Emergency Action Plan with Emergency contact numbers, Employees' name who are trained on CPR & first aid to be displayed in Electrical rooms.
- 8) First aid boxes, stretchers, Oxy-pack, Gas Mask etc to be made available in Electrical rooms or control rooms.
- 9) The Electrical Installations for high and extra high voltage shall have a running tap of drinking water and Emergency Safety shower with drinking water connection and proper drainage facility. In case of electric burns, the burnt portion should be cooled down using drinking water. In every manned high and extra high voltage sub section, an artificial respirator to be provided.
- 10) Good housekeeping to be maintained within the electrical room. The floor of Electrical Substation should be kept free from slip/ trip/ fall hazard like open cable riser, moving checker plate or floorboard etc.
- 11) Along with the main exit door, Emergency Exit door is to be provided and should be in accessible safe condition in the 415V and above electrical rooms.
- 12) When dead mains are connected to live mains, all connections to the live parts shall be made last and in all cases the phases sequence should be checked to ensure that only like phases are connected together by testing Phase Sequence tester Rod & Phase Sequence Meter for HT & LT respectively.
- 13) Only Authorized person is authorized to work on live low and medium voltage mains and apparatus with adequate precautions & PTW system. For earthing and short-circuiting only appropriate methods (earthing chains, earthing rods etc.) should be

- used. Measures shall be taken against, the inadvertent energizing (back charging) of the mains and the apparatus.
- 14) All disconnect switches are to be identified and labelled stating the source of the circuit and the equipment / circuit it is feeding.
 - 15) Electrical equipment, machinery and work areas to be kept clean. Access to panels and junction boxes to be kept clear.
 - 16) Insulation mat of proper grade to be used in front of 415V & above electrical switchgears.
 - 17) All panel doors must be locked, made vermin proof by sealing entry holes specially from cable entry points.
 - 18) There should not be any loose wirings, exposed joints, Distribution Box or Junction box in cover open condition in the workplace.
 - 19) No High Voltage and Extra High Voltage system/switchgear which has been kept disconnected, for period of six months or more for any maintenance job shall not be energized without Testing.
 - 20) All apparatus, cables and supply lines shall be maintained in healthy conditions and tests shall be carried out periodically as per the relevant code of practice of the Bureau of Indian Standards.
 - 21) Records of all Tests, repairs, maintenance work of all the Equipment in the Electrical room shall be duly kept in the history registers, so that these records can be compared with earlier ones.
 - 22) Only authorized /qualified personnel to be allowed to work on electrical job. Relevant certification to be obtained from Electrical Inspectorate as applicable.
 - 23) No one should wear loose clothing, metal watch straps, bangles or finger rings while working on electrical appliances.
 - 24) Any electrical appliance or machinery never to be used by user while in wet condition.
 - 25) The tools & tackles like screw drivers, pliers, cleaning brushes, Handlamps, torches etc should be properly insulated.
 - 26) Cords or tools with worn insulation or exposed wires should not be used.
 - 27) Only double-sided steps or stable ladders of FRP (dual support) to be used for maintenance on panels and equipment.
 - 28) Before any maintenance job is carried out on or near the live parts the power source should be proved dead. All voltage indicators should always be tested on a known "Live" source immediately before and after use. It is recommended to use non-contact type voltage detector after isolation & before starting any work in Switchyard & Switchgears.
 - 29) In no circumstances, attempts should be made to rewire a non-rewire able fuse.
 - 30) Area is to be guarded off wherever possible, where men are working on H.T. equipment.
 - 31) Switchyard Hot line maintenance activities are not recommended in any case though the work is of emergency nature. All maintenance activities are to be carried out after proving dead of the element. However, Insulator (BPI, Disc Insulators & CTs) hotline washing may be carried out by authorised agency only with considering all the safety aspects.
 - 32) While working with welding machine, following safety practices to be adhered:
 - Worker must have license for welding work.
 - ELCB must be installed in power supply circuit.
 - Be sure the welder is properly installed and grounded.

- Take proper precautions to prevent fires. Fire extinguisher must be available.
- Wear eye protection always.
- Weld only in a firesafe area.
- Never do any welding, cutting, or hot work on used drums, barrels, tanks, or other containers.
- Keep a well-stocked first aid kit handy.

Definitions:

- Low Voltage (LV): Where the voltage(rms) does not exceed 1KV under normal conditions.
- Medium Voltage (MV): Where voltage(rms) range in between 1KV & 33KV under normal conditions. ($1KV < U_n \leq 33KV$)
- High Voltage (HV): Where voltage(rms) range in between 33KV & 150KV under normal conditions. ($33KV < U_n \leq 150KV$)
- Extra High Voltage (EHV): Where the voltage exceeds 150 KV under normal conditions.
- PPE- Personal Protective Equipment
- ATPV- Arc Thermal Performance Value

10.0 SAFETY INSTRUCTIONS FOR WORKING ON ELECTRICAL EQUIPMENTS:

10.1 Minimum electrical clearances as per standard:

No person shall work within minimum working distance from the exposed live mains and apparatus. The minimum working distance depends upon the actual voltages. Exposed live equipment in the vicinity shall be guarded off in such a way that it does not hinder the movement of the maintenance personnel.

For indoor system (Switchgear & Main Plant):

Voltage in kV	Phase to earth in mm	Phase to phase in mm
0.415	15.8	19.05
3.3	50.8	
11	76.2	127.0
33	222.25	355.6

For outdoor system (Switchyard):

Voltage in kV	Phase to earth in mm	Phase to phase in mm	Safety working clearance (mm)	Ground clearance (mm)	Height of Insulator bottom from Ground (mm)
33	320		3000	3700	2440
400	3500	4200	6500	8000	

Note:

- Phase to phase clearance is the minimum clearance required to be provided between the live parts of the different phases.
- Phase to earth clearance is the minimum clearance required to be provided between the live parts of the system & earth structures.
- Safety working clearance is the distance between the live parts of the phases & the terminal of the work section. Work section or Maintenance section may be a platform or ground on which a maintenance personnel can carry out his task safely.
- Ground clearance is the vertical clearance between the live parts of the equipment & the earth surface.
- It is the height of the lowest portion of Insulators used for supporting of live conductors.

10.2 Operations of Switches and Isolators

- 1) No high voltage switch, isolator or earthing switch shall be operated or earth connection attached or removed without the sanction of an authorized person, except in the case of moveable earth connection (temporary earthing) on high voltage overhead lines, which may be fixed or moved by an authorized person under the direction of the permit to work, which authorizes him to carry out the work.
- 2) When a switching operation to be carried out, the authorized person shall convey his instructions to the operating person in detailed to carry out the operations.
- 3) All the switching operations (Isolation/Normalization) has to be carried out as per station SOP which should be checked & reviewed before any operation. Switching operation at Switchyard to be carried out in presence of Electrical Maintenance person.
- 4) During switching, message to be clearly delivered between authorized person & operators & all communication should be recorded in register.

10.3 Testing of mains and equipment:

- 1) No person shall apply test voltage to any mains unless he has received a permit to work and has warned all persons working on the mains of the proposed application of the test voltage. If any part, which will thus become alive is exposed, the person in charge of the test shall take due precautions to ensure that the exposed live portion does not constitute danger to any person. It should also be ensured before the application of test voltage, that no other permit to work has been issued for working on this main.
- 2) When equipment is isolated from the mains supply for the testing, the official responsible may give sanction for the operation of switches, isolators, earthing switches, earth connections etc. and for the application of testing supplies to the isolated section, without reference to him. The person in charge of the testing then becomes wholly responsible for the safety precautions within the isolated sections but no switch or isolator connecting any isolated sections to the main supply system shall be operated without direct sanction of the responsible official except for purpose of obtaining testing supplies.

- 3) During testing of any equipment, all testing devices to be firmly earth before making any connection with the equipment.

10.4 Working on cables:

10.4.1 Identification of cables to be worked upon

A cable shall be identified as that having been proved dead prior to cutting or carrying out any operation which may involve work on or movement of the cable. A non-contact indicating rod, induction testing set may be used for proving the cable dead for 1 phase Un-armoured cables.

10.4.2 Working on High Voltage Cable

Work on high voltage cables shall be only permitted on receipt of the permit to work. In addition to the precautions taken under the person carrying out such work shall be personally instructed on the spot by an authorized person who shall first satisfy himself that the cable has been made dead isolated and earthed and if possible, the switch controlling the cable drawn from the cubicle and suitable danger boards installed in position.

10.4.3 Working on underground cables

- 1) For isolation of cables, open at least one set of disconnecting switches or fuses in every source through which the cables can be made alive including leads to the cable of potential transformers and then discharge the cable to earth.
- 2) Cable route indicators should be provided (if approved drawing is available) and cable route records maintained. It would access the particulars of all underground cables correctly in the vicinity of the faulty cable.
- 3) Use of sharp-edged crowbars or pick-axes should be avoided during excavation while locating the faulty cable or laying new cable.
- 4) All the cables in the vicinity in the fault area shall be exposed and identified to establish the identification of the faulty cable.
- 5) Before any high voltage joint of chamber is to be opened in circumstances where it is not desirable to spike the cables or earthing the joint or chambers, the authorized person shall satisfy from cable route record and, if necessary, by approved tests that the joint or chamber is associated with the particular cable which has been made dead and it is safe to work on it.
- 6) Employees shall not step on live cables even though those are insulated and enclosed in a lead sheath. Tools and materials shall not be rested against the sheath of the cable.

10.5 Working on Height

Before any work is began on any pole or tower of a high voltage overhead line, which is adjacent and parallel to any other high voltage overhead line with conductors “alive” or any pole or tower which supports, more than one set of high voltage conductors “alive” the following special precautions, in addition to the foregoing, shall be taken in every case:

- 1) The authorized person in charge of the work shall ensure that each workman who is to work on the poles or towers is definitely informed and thoroughly

understands on which set of conductors or equipment the work is to be carried out.

- 2) A “red” flag / Caution Tape (or lamp at night) which are available with maintenance crew, shall be displayed on the side of the pole or structure on which the conductors are “alive”.
- 3) Work shall not be performed on any higher position of tower / line when a line below is energized.
- 4) Proper height permit to be taken from operation & arrangement at site including PPEs are to be verified by safety dept. also. Work to be started only after clearance from safety dept. person.

10.6 Working on High Voltage Equipment and Overhead Lines

Before commencing any work of repairs, alterations, extensions, additions or cleaning of high voltage apparatus, the following operations shall be carried out in sequence:

- 1) The apparatus or cable or transmission line shall be switched out and isolated from all points of supply under the direction of the authorized person.
- 2) The switches, isolators and control links shall be locked in position by the keys provided for the purpose.
- 3) Safety Tags (as per PTW guidelines) shall be placed at all points where apparatus can be made alive.
- 4) All apparatus shall be discharged to earth and efficiently connected to earth near all points from which supply could be connected to it or between such points and the place of the work. All earthing shall be done by the approved methods. The earthing leads used for earthing shall be of adequate cross section according to voltage levels to enable passage of the fault current without fusing.
- 5) Earthing shall also be carried out at the point of work by means of temporary earths on each phase and in no case shall the temporary earths be removed from two phases simultaneously while the work is being carried out.

10.7 Working on deadlines and equipment:

10.7.1 Grounding of Lines and Equipment

- 1) Before doing any work on deadlines or equipment where there is a possibility of their becoming energized from any source, such line or equipment should be short-circuited and grounded between the location of work and all possible sources of energy.
- 2) Conductors to be grounded should be checked for potential by an approved method before the ground is installed.
- 3) Temporary grounding cables shall be flexible stranded copper not less than No.10 and shall be equipped with approved clamps at each end.
- 4) Grounding cables should be inspected before each use.
- 5) When grounding lines or equipment, the connection to the ground shall be made first and that to the circuit or equipment last. In removing grounds, first remove the connection to the circuit or apparatus and then remove the ground connection. Insulated hot sticks should be used in making the ground connection to the circuit or apparatus.

- 6) Grounds shall be placed on all phases even if work is to be carried out on one phase only.
- 7) For work on the line, ground shall be placed at nearest tower on each side of the point of work, but in no case should earths be more than six spans apart. As an additional safety measure, if possible, in addition to above grounds, line should also be grounded on the tower where the work is to be carried out.
- 8) When work is to be carried out on lines of all-insulated construction and grounding point is not provided at point of work, temporary grounds shall be connected at point of work to an efficient portable earth stake driven into the ground. The line shall also be grounded at the nearest line grounding point on either side of the point of work.
- 9) Where two or more crews are working independently on the same line or equipment, each crew shall properly protect themselves by placing their own temporary grounds.

10.7.2 Line work on poles and towers

- 1) Before climbing any elevated structure, every employee shall first assure himself that the structure is strong enough to sustain his weight safely.
- 2) If poles or cross arms are apparently unsafe because of decay or unbalanced tensions of wires on them, they shall be properly braced or guyed before they are climbed.
- 3) Linemen shall wear their safety lines while working on the poles and towers.
- 4) Wire hooks shall not be attached to linemen's belts.
- 5) Safety straps should not be placed above the top cross arm when it is at the top of the pole.

10.7.3 Working on lines under adverse weather conditions

In the event of the near approach of lightning or thunderstorm all work on overhead lines shall cease immediately.

10.8 Working on switching operations

- 1) Every message relating to the switching operations on the high voltage system shall, wherever practicable, be written down. Every such message shall be repeated in full to the sender to ensure that the message has been accurately received (as per PTW guidelines).
- 2) A record of high voltage switching will be entered in station log.
- 3) All breakers and isolators should bear lettering or sign boards to indicate the circuit they control.
- 4) When releasing the electric circuits, breakers or equipment for work on them, the associated breaker and disconnecting switches shall be opened in the following order:
- 5) -The breaker will be opened first.
- 6) -The isolator will be opened, but before operating the isolator, it shall be made sure that the breaker is open.
- 7) After opening isolators and air-break switches, check carefully to see that all blades are in full open position.

- 8) When lines and circuits are taken out of service, the breaker control circuit should be opened either by operating the opening device or by removing the control circuit fuses.
- 9) If the circuit is controlled by automatic re-closing breaker, the re-closing mechanism shall be made inoperative.
- 10) Isolators shall be closed in firm positive manner, using sufficient force to make full contact of blades.
- 11) Before removing fuses, switches should be opened if provided. Removing fuses from inductive circuit carrying current without opening the switch is hazardous.

10.9 Working on transformer

- 1) When work is to be carried out on a transformer, both low and high tension breakers and isolators shall be opened. Similarly, during isolation of transformers to which potential transformers are connected, such potential transformers shall be isolated.
- 2) Before starting any work on a transformer installation, it is important to check carefully for back feed, abnormal voltage or other dangerous conditions. Unusual circuit conditions may exist which require special consideration.
- 3) Whenever transformers are replaced, the new transformer should be checked carefully for voltage, polarity and phase sequence before taking into service.
- 4) Area should always be cordoned off & Safety tagging should be done prior to starting the job on transformer.
- 5) Spare oil shall not be stored in the vicinity of any oil filled equipment in any such substation or switching station.
- 6) The transformers of 10 MVA and above rating or in case of oil filled transformers with oil capacity of more than 2000 litres are to be provided with firefighting system.

10.10 Working on instrument transformers

- 1) The base body of all instrument transformers should be grounded.
- 2) Current transformers secondaries should never be open circuited when current is flowing in the primary.
- 3) The secondary circuit of current transformers should be connected to ground at all times when the transformer is in service.
- 4) Potential transformers secondaries should never be shorted.
- 5) The low voltage winding of potential transformers should always have one side permanently and effectively grounded.

10.11 Working on filtration of oil of transformer

When carrying out work of filtering of oil on transformers, care shall be exercised that all exposed live conductors are suitably barricaded so that no person and no apparatus such as flexible hose etc. that is being handled comes in contact with the live parts. All such work shall be carried out under the direct supervision of an authorized person.

10.12 Working on switch gear and control panel

- 1) While working on manually operated panel mounted circuit breakers when the operating handle is on the front and the circuit breaker is on the rear of switchgear or on another panel, a danger notice shall be placed on the handle.
- 2) When the work is to be carried out on the bus bars spouts the following operations shall be carried out.
- 3) -The section of bus bars on which the work is to be carried out shall be made dead and shall be isolated from all points of supply.
- 4) -The isolating arrangements and the shutters of live spouts shall be locked so that they cannot be operated.
- 5) The bus bar shall be earthed with approved earthing equipment at a panel other than at which work is to be done and the isolated section of the bus bars.
- 6) Bus coupler panel and its cubicle shall be handled with utmost care. Red painting of Bus coupler, Adaptor panel & Tie feeder panel from all sides (front, top and back) for easy identification and treat Adaptor panel as a part of Bus coupler.
- 7) Incoming barrier of power supply and phase barrier inside the switchgear panel to be ensured.
- 8) There should not be any loose wirings, exposed joints, Distribution Box or Junction box in cover open condition in the workplace.
- 9) During inspection or maintenance of mechanism of Circuit breaker care should be taken to avoid fingers being trapped in any part of the mechanism and avoid the possibility of anyone being struck by moving parts of the mechanism or the moving contacts. It is recommended to discharge the stored energy in the mechanism like release of the spring etc.

10.13 Working on storage batteries

- 1) When making electrolyte for storage batteries always pour acid into the water. The reverse method may cause an explosion. Suitable goggles or face shields should always be worn when making electrolyte. Ensuring the usage of PPE's by outsourcing staff is the responsibility of the Shift Officer.
- 2) Smoking and use of matches or other open flames are not permitted in battery rooms or while inspecting filling, testing or handling batteries.
- 3) Water connection & wash basin must be ensured in Battery room. Portable eye shower may be used where water connection is not available.

10.14 Working in switchyard (ais) for carrying out shutdown work & testing:

The single Line diagram of the switchyard mentioning the detail nomenclature of the bay equipment must be displayed at the entrance of the control room and must be available in the switchyard control room for reference. Display Board is to be placed with Dos and Don'ts at the entry of the switchyard.

Bay wise Isolation and Normalization procedure for all voltage levels of the station is to be prepared and converted in the form of LMI. It is to be ensured that the document is available in the Switchyard Control Room.

- 1) Phase Identification (R/Y/B) of the all the bay equipment /Bus post insulators is to be done either by paint/stickers (Red for R phase, Yellow for Y phase & Blue for B phase).
- 2) Before commencement of any shut down work or testing in an electric supply line or apparatus, the Engineer or Supervisor in-charge of the work or testing shall identify the possible hazards, such as; electrocution, flash over, fall of person from height, fall of objects from height, failure of Tools & Plants, fire, etc., that may be encountered while carrying out the work or testing near charged area and take necessary precaution to protect the working personnel.
- 3) The Engineer or Supervisor in-charge of the work shall, before commencement of any work, brief the entire working group or gang of the hazards that may be encountered and the necessary precautions to be taken by them.
- 4) The Engineer or Supervisor in-charge of the work shall obtain proper Permit-To-Work(PTW) from the concerned Operation In-charge(s) and ensure that the electric supply line or apparatus or section is isolated from all sources of energy, de-energized and earthed.
- 5) The Engineer or Supervisor in-charge of the work shall ensure that adequate and Appropriate local earths are fixed at the zone of working, and the earthing rods remain connected to the isolated section of the electric supply line or apparatus or section till all men and materials have been moved away to safe zone and PTW is returned on completion of the work.
- 6) If the local earths are required to be removed for any testing purpose, the same shall be done only when all the working personnel are in the safe zone, on the ground or on the tower, and in the presence of the Engineer or Supervisor. If the working personnel are required to go up or approach the conductor(s) subsequently for any work, such as, removal of test leads, tightening or adjustment, they shall be permitted to proceed only after re-fixing the local earths, as required.
- 7) The Engineer or Supervisor in-charge of the work shall positively confirm by suitable means that the electric supply line or apparatus or section is totally dead before giving clearance for the working personnel to approach same.
- 8) The Engineer or Supervisor in-charge of the work shall, while carrying out the shutdown work or testing, ensure that working personnel are maintaining safe distance from the adjacent charged electric supply line or apparatus or section, and also, no objects, such as, Tools & Plants, ladders, cranes, man-lifts, etc., are moved, so as to infringe the safe distance, endangering the working personnel.
- 9) Mobile cranes, derricks, manlifts and wheel mounted ladders shall be effectively earthed when being moved or operated in close proximity with energized apparatus or section.
- 10) Portable ladders and poles shall be carried only in the horizontal position when being moved in close proximity with energized lines or equipment or area.
- 11) Adequate and effective supervision shall be ensured by the owner as well as the contractor for all activities while working or testing on electric supply lines and apparatus when any shut down work or testing is done near charged electric supply line or apparatus or section.
- 12) Lone worker shall never be allowed to work on electric supply lines, equipment and apparatus or while testing.

- 13) Sufficient supervisory personnel shall be deployed for close monitoring while various type of works are under progress at the same or different locations. Supervising work shall never be delegated to the subcontractors' personnel.
- 14) The deployed Supervising Personnel shall not leave the working spot when shut down work at height or testing is in the progress, as the working personnel may not be aware of the consequences of unsafe practices. No other work, which requires them to move out of the location, shall be undertaken by Supervising personnel, when shutdown work or testing is in the progress.
- 15) Wherever shut down activities are required to be carried out for more than one day on the any electric supply lines, apparatus or section, earthing(s) provided at the said work site shall be inspected by the Engineer or Supervisor every day morning for their healthiness, fitness and proper tightening, before giving clearance for the working personnel to climb the tower or structure to resume the work.
- 16) Gates in switchyard fences and doors to switch gear and other enclosures containing live equipment, or other hazards, should be kept locked at all times except when Authorized Person entered for working inside.
- 17) When carrying ladders, pipes, conduits, reinforced rods and other long material in to stations, switchyards, switch gear rooms and other places where there is a danger of touching the live parts, the material should be held by two men, one at each end, and carried in the hands and not on the shoulders.
- 18) Isolators/switches closing on the section of bus bars on which work is to be carried out shall be locked in open position and the closing mechanism rendered inoperative.
- 19) While working on the outdoor structure at a height more than 1 meters from the ground level, safety equipment such as safety belts, handling, etc. should be used.
- 20) No person shall stand directly below the place of work when the work is in progress in the outdoor
- 21) structure to avoid any tools or bolts or nuts or clamps etc. falling on their heads.
- 22) Helmets should be invariably used while working on the outdoor structures, on the outdoor structures, both by the men stationed at the ground and those on the structures.
- 23) Isolators and controlling circuit breaker shall be interlocked so that isolators cannot be operated unless the corresponding breaker is in open position.
- 24) Isolator and corresponding earthing switches shall be interlocked so that no earthing switch can be closed until the corresponding, isolator is in open condition.
- 25) Where two or more supplies are not intended to be operated in parallel, the respective circuit breakers or linked switches controlling the supplies shall be interlocked (Electrically and mechanically to extent possible) to prevent possibility or any inadvertent paralleling or back feed.
- 26) When working in the vicinity of circuit breakers or buses use every precaution to avoid injury from arcing.
- 27) It is to be ensured that there is no accumulation of dry grass/shrubs/herbs (which is a potential threat for fire hazards) inside and around the switchyard including the cable trenches.

10.15 Working in switchyard (GIS) for carrying out shutdown work & testing:

- Not Applicable –

10.16 Safety practices during O&M of Solar power plant:

While solar energy is among the cleanest and least hazardous of energy sources, there are four significant hazards during installation and maintenance: Lifting, Trips & Falls, Electrical, and Ladders.

10.16.1 Safety practices for lifting:

- 1) Use mobile carts, forklifts, or other non-manual methods to reduce lifting.
- 2) Use gloves to protect your hands and improve your grip.
- 3) Get a secure grip with both hands.
- 4) Keep the load as close to the body as possible.

10.16.2 Safety practices for trips & falls:

- 1) Identify all potential slip, trip, and fall hazards before starting.
- 2) Keep work areas clear of debris and obstructions, including loose shingles, poorly placed tools, and electrical cords.
- 3) Ensure work surfaces are free from ice, oil, water, and other substances.

10.16.3 Safety practices for electrical:

- 1) Inspect your equipment to ensure it's in safe working condition.
- 2) Work a safe distance from power lines.
- 3) Cover the solar array with an opaque sheet to "turn off" the sun's light.
- 4) Always test circuits to ensure they are de-energized before working on them.
- 5) Use a current clamp to check for hazardous energy before working on PV panels.
- 6) Take special care around inverters. These can hold a powerful charge even when power is removed.
- 7) Never disconnect PV connectors or other equipment that's under load.
- 8) Wear the appropriate PPE for electrical safety.

10.16.4 Safety practices for ladders:

- 1) Inspect ladders before use. Mark defective ladders with a label stating "Do Not Use."
- 2) Use a fiberglass ladder with non-conductive side rails near power sources. Aluminium and metal ladders are hazardous near power lines or electrical work.
- 3) Ensure that ladders will extend a minimum of three feet above the last rung that the worker will stand upon.
- 4) Place the ladder on dry, level ground, away from walkways and doorways, and a safe distance from power lines. Secure ladders to ground or rooftop.
- 5) Grasp the horizontal ladder rungs (not the vertical rails) and maintain 3 points of contact.
- 6) Never carry a solar panel or other equipment while climbing a ladder. Use a winch or hoist system to lift solar panels to the roof.

10.16.5 Personal protective equipment (PPE) for solar workers:

Personal protective equipment (PPE) is essential during every solar installation. Employers have to assess workplaces for hazards and make sure they provide workers with the necessary PPE for their safety. This can include:

- Safety helmet
- Eye protection, like glasses or goggles
- Vests
- Harnesses
- Safety nets
- Personal fall arrest systems

10.16.6 Additional safety requirements for solar installations:

- 1) Proper Segregation and clearance to be provided between positive and negative components including wiring, terminal boxes, etc. inside the combiner box.
- 2) While working on the combiner box, it shall be ensured that there is no current flow and is completely isolated from input side.
- 3) Clear pathways of minimum 75 cm in width for roof access and emergency exit shall be provided for roof top system. There should be clear pathways, walkways between the rows/columns of solar panels as is required for cleaning and maintenance. Cables should be laid in trenches as it would obstruct the movement of manpower and cleaning equipment.
- 4) Earthing for AC and DC system shall be interconnected to have equipotential bonding.
- 5) Protective earthing shall be provided inside the inverter cabinet.
- 6) All metal casing/shielding of the plant, each array structure of the PV yard, equipment, inverters & control systems shall be grounded through proper earthing.
- 7) Portable fire extinguishers shall also be kept at various locations in plants to take immediate action in case of fire caused by electrical short circuits.

10.17 Electrical earthing requirement:

- 1) Equipment Earthing: The frame of every generator, stationary motor, portable motor, and the metallic parts (not intended as conductors) of all transformers and any other apparatus used for regulating or controlling energy and energy consuming apparatus shall be earthed by two separate and distinct connections with earth.
- 2) All earthing systems shall be tested for resistance with the help of an "Earth Tester" on a dry day during the dry season, not less than once a year for high and extra high voltages (>650 Volts) and not less than once every two years for low, medium voltages (<=650 Volts). The combined value of resistance to earth should not exceed 1 Ohm for all voltages.
- 3) The earth pit identification number, latest earth resistance, date of testing and due date of testing shall be displayed on every earth pit name plate.
- 4) Earth pits to be properly maintained and chemical treatment of soil to be done as per requirement to keep the earth resistance to a satisfactory value.
- 5) Earth pits should be properly covered.

- 6) No material shall be allowed to be stacked/ dumped/ accumulate and debris shall not be deposited near/within the earthing station area.

10.18 Electrical fire safety:

- 1) Any combustible materials like grease, paint etc should not be present in proximity with live electrical equipment.
- 2) Fire buckets with clean dry sand and ready for immediate use for extinguishing fires in addition to fire extinguishers suitable for dealing with electrical fires (CO₂ or DCP), shall be conspicuously marked and kept in all generating stations, enclosed substations and switching stations in convenient location.
- 3) Fire alarms to be provided in all Electrical rooms, man approachable cable vaults, cable tunnels, analyser rooms, PRS rooms and its hooter are sounded at a 24x7 manned area like control room, shift office etc and checked on regular interval.
- 4) Fire Retardant painting to be provided on Electrical cables to stop propagation of fire.
- 5) Training to be provided on Fire Safety & CPR to electrical personnel.
- 6) Regular mock drill to be conducted for total power failure, electrocution, electrical fire etc as per the electrical area's emergency preparedness plan and record to be maintained.
- 7) Wastepaper, rags and other combustible material should not be allowed to accumulate.
- 8) Flammable liquids shall be kept in approved safety cans and identified by proper labels.
- 9) Varnish, paints, lacquers and thinners are highly inflammable and should be stored away from all open flames or possible sources of ignition. Matches and open flames should not be used where varnish paint or lacquer is being applied with a spray gun.
- 10) Open flames and smoking are prohibited in all areas where inflammable liquids or gases are stored or being used. Such areas shall be posted with appropriate warning signs.
- 11) All employees should be familiar with the location and proper use of fire extinguishers in their work area.
- 12) No employee should smoke or use matches or open flames on customer's premises unless it is positively known that such action do not conflict with the customer's rules.

10.19 Rescue & first aid measures in electrical shock:

- 1) The victim usually gets stuck to the source of the electricity, and it is important that the victim should be first separated from the electrical source.
- 2) The main power supply switch to be adequately turned off.
- 3) In certain circumstances, it may be quicker to simply pull the victim away from the electrical source.
- 4) The victim should not be touched with bare hands, or the electric current will passthrough the rescuer as well. Some dry, nonconductive material such as a

- wood, insulated rod etc. to be used to separate the victim from the live current whatever is handy.
- 5) Once the victim has been separated, it is to be checked to see if he is breathing.
 - 6) If breathing has stopped or seems slow, CPR (Cardio-Pulmonary Resuscitation) to be administered immediately by a trained person.
 - 7) The victim should be moved as little as possible because the person may have suffered injuries to spine and neck.
 - 8) If the victim has an electric burn, the clothing from the burned area to be removed gently (unless it's stuck to the skin) and the burned area to be rinsed in cool, running drinking water.
 - 9) Emergency medical attention to be called as soon as possible.

First Aid Instructions: -

- 1) Remove the patient from the source of accident / remove the cause of injury.
- 2) Keeps the injured person lying down in a comfortable position.
- 3) If the breathing has ceased, immediate measures must be taken to restore it.
- 4) If the patient has received burns attend to them.
- 5) When the patient has fractured a bone, no attempt must be made to move him.
- 6) Treat the patient for shock.
- 7) Send for medical help.
- 8) Never give water to patient.
- 9) Keep by standards away from the patient.
- 10) Keep the patient warm.

10.20 Housekeeping:

- 1) Good housekeeping to be maintained within the electrical room. The floor of Electrical Substation should be kept free from slip/ trip/ fall hazard like open cable riser, moving checker plate or floorboard etc.
- 2) Workmen are frequently injured, by stumbling, stepping on, or bumping into tools, material and other objects left lying around, or by objects falling from above.

To ensure good housekeeping, following precautions should be observed: -

- 1) Walks, stairways, fire escapes and all other passageways shall be kept clear of all obstructions.
- 2) Tools and materials should not be placed where they may cause tripping or stumbling hazards or where they may fall and strike anyone below.
- 3) Puddles of oil and water create slipping hazards and should be cleaned up promptly.
- 4) Nails in boards, such as those removed from scaffolds, forms and packing boxes, constitute hazards and should be removed. The boards should be carefully stacked or stored.
- 6) Dirty and oily waste rags should be deposited in approved containers and disposed off as soon as practicable to avoid fire hazard.
- 7) Broken light bulbs, glass metal and scrap and other sharp objects should be dumped in places or containers provided specially for them.
- 8) Discarded fluorescent and other gas filled tubes shall be disposed off safely.

- 9) Places where persons work or pass in emergencies, shall be provided during time of use with adequate lighting (natural / artificial / or both) for operations or special type of work performed.
- 10) General lighting shall be of a uniform level widely distributed.
- 11) In big installations / offices emergency lighting shall be provided.
- 12) Adequate ventilation shall be provided in workplaces by natural / artificial means.

10.21 Personnel Protective Equipment (PPE's) and devices:

The company shall provide adequate & approved PPE's for various jobs depending upon the hazard. The centralized procurement of PPE's shall be made which will ensure the quality as per respective standards.

PPE guidelines: -

- 1) Use suitable protective equipment, like Arc flash suite, Insulation gloves, safety helmet, safety shoes, safety belts, Insulation mats, safety glasses, etc., wherever required as per instructions or wherever it provides greater safety.
- 2) All safety devices should be checked before starting work.
- 3) Safety equipment should be tested at frequent intervals to ensure that equipment would provide the safety desired.
- 4) Protective gears such as helmets, safety shoes, safety belts are issued to linesmen, jointers, supervisors as applicable for personal protection and their usage is monitored.
- 5) It is responsibility of supervisor to ensure the usage of P.P.E.'s.
- 6) The P.P.E.'s shall not be carried / stored with tools etc. to avoid damage to them.
- 7) Any employee working on height above 8 feet from ground except working on platform should use Safety Belts.
- 8) Uses only approve type of operating rods.
- 9) Operating Rods shall be kept as dry as possible. It should not be dropped / left lying on ground.

Hand Tools: -

- 1) Many accidents result from improper use of tools and use of defective tools and equipment.
- 2) Employees should use only those tools and equipment, which are in good condition, and only of the purpose for which they are designed. Where proper and safe tools are not available for the work at hand, then employee shall report the fact to his supervisor.
- 3) Tools, which develop defects while in use, should be removed from the service, tagged and not used again until brought in good condition.
- 4) Impact tools such as chisels, drills, hammers and wedges with mushrooms heads should not be used until they have been reconditioned.
- 5) Hammers, axes, shovel and similar tools should not be used if handles are loose, cracked or splintered.
- 6) Defective wrenches such as open end and adjustable wrenches with spread jaws or pipe wrenches with dull teeth should not be used, as they are likely to slip.
- 7) Pipe or other extensions should not be used on a wrench handle to increase the leverage unless the wrench is specifically designed for such an extension.
- 8) Portable electric tools preferably should be equipped with 3-wire cord having the ground wire.

- 9) connected to the tool frame and is to be grounded at the other end.
- 10) Metal rules, metal tape lines or lines containing wires shall not be used around electric conductors or equipment.
- 11) All tools carried on trucks should be inspected each month and defective tools repaired or replaced.

Ladders: -

- 1) Inspect the ladder before use.
- 2) Ensure firm footing.
- 3) Secure at top or have a man at the foot.
- 4) Ensure correct angle (75 degree) or position ladder 1 foot out at base for every 4 feet of vertical height.
- 5) Ensure that the ladder rises 3 feet above landing point.
- 6) Face the ladder when climbing or descending.
- 7) Avoid make shift arrangement in lieu of ladder.
- 8) Ensure the rubber shoes at both arms and at each terminal. i.e. at each end.
- 9) Before fixing confirm that no electrically charged conductor is passing nearby.

Protective barrier (temporary): -

- 1) When the work is conducted along public streets or highways, pedestrian and vehicular traffic shall be warned by signs and flags by day and red lights or flares by night. Wherever necessary, signalmen should be provided.

Earthing Devices: -

- 1) Only approved earthing devices shall be used in all work.
- 2) Care shall be taken to maintain earthing by ensuring condition of clamps.
- 3) Non-contact type live line indicator rod may be used to check the bus bar area, cables, overhead conductors but prior to use it should be ensured that indicator rod is in good condition.

11.0 SAFETY CHECKLIST BEFORE WORK EXECUTION IN COMMON ELECTRICAL AREAS:

Equipment/Feeder description:

Date:

SI No.	Activity	Responsibility	Yes/No
1	Ensure submission of JSA & SWMS for work	EMD	
2	Ensure issuance of proper PTW through SAP	Operation	
3	Physical verification of tools & tackles, earthing rods, ladders, hot line sticks, all required PPEs etc.	EMD	
4	Ensure Availability of first aid box, fire extinguisher, water etc. as per nature of work.	Oprn & EMD	
5	Ensure Isolation done as per SOPs mentioned.	Oprn & EMD	
6	Barricading of working area by safety tag	EMD	
7	Pep talk arranged for the deputed workers	EMD	
8	Ensure the use of proper PPEs as per work condition by all workers	EMD	
9	Ensure the equipment is de-energized by use of hot line stick or multimeter. Then apply earthing rod before start of work (If applicable). Ladder/vehicles must be temporarily earthed.	EMD	
10	Always Maintain the safe electrical clearance as per clause 7.1	EMD	
11	After completion of work remove temporary earthing. Remove all person & materials	EMD	
12	Returning of PTW & Ensure normalization done as per SOPs.	Oprn & EMD	
13	Physical monitoring of equipment & area after system normalization	Oprn & EMD	

- Note: For Switchyard & Switchgear system, Equipment Operation (Isolation/Normalization) must be adhered as per CC-OS guidelines vide COS-ELEC-OGN-041, COS-ELEC-OGN-042, COS-SYS-OGN-07 etc.
